Project Report

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Subject Name: Internet of Thing **Subject Code: 24-CAP-751**

Aim: To develop a smart energy meter using ESP32 that measures and monitors electrical energy consumption in real time and allows remote control of appliances using the Blynk IoT platform.

Objectives:

1. To measure electrical parameters
    - Measure voltage using the ZMPT101B Voltage Sensor Module and current using ACS712 sensor.
2. To calculate power and energy consumption
    - Compute real-time power and total energy usage.
3. To implement real-time monitoring system
    - Display live data using the ESP32 Development Board.
4. To enable IoT-based remote monitoring and control
    - Use the Blynk to monitor data and control appliances remotely.
5. To control and protect electrical loads
    - Use relay module to switch loads ON/OFF and handle overload conditions.

Components Required:

Hardware

1. **ESP32 Development Board**
   - 32-bit powerful microcontroller
   - Built-in Wi-Fi & Bluetooth
   - Reads sensor data and controls relay
2. **ZMPT101B Voltage Sensor Module**
   - Measures AC voltage
   - Provides analog output proportional to voltage
   - High accuracy with isolation
3. **ACS712 Current Sensor Module**
   - Measures AC current
   - Outputs analog signal to ESP32
4. **Relay Module**
   - Controls AC load (bulb)
   - Electrically isolates high voltage
5. **Breadboard**
   - Temporary connections
6. **Jumper Wires**
   - Circuit wiring
7. **AC Bulb (Load)**
8. **Power Supply (5V USB)**

Software Components

- Arduino IDE (for programming)
- Embedded C / Arduino code
- Optional: IoT platform (like Blynk)

Details of Components:

1. **ESP32 Development Board**
   ESP32 is the main controller of the system. It reads data from voltage and current sensors, processes the values, calculates power and energy, and sends data to the mobile app. It also controls the relay for switching the load.

2. **ZMPT101B Voltage Sensor Module**
   This sensor is used to measure AC voltage. It provides an analog output proportional to the input voltage and ensures electrical isolation for safe operation.

3. **ACS712 Current Sensor**
   It measures the current flowing through the load using the Hall effect principle. The sensor outputs an analog signal which is read by the ESP32 for current calculation.

4. **Relay Module**
   The relay acts as a switch to control the AC load. It receives signals from the ESP32 to turn the load ON or OFF.

5. **Blynk**
   Blynk is used for IoT integration. It displays real-time values like voltage, current, and power, and allows the user to control the load remotely.

6. **Power Supply**
   A regulated power supply is used to provide the required voltage to the ESP32 and other components.

7. **Load (Bulb)**
   The bulb is used as a test load to measure energy consumption and demonstrate the working of the system.

Block Diagram of Designed Model:

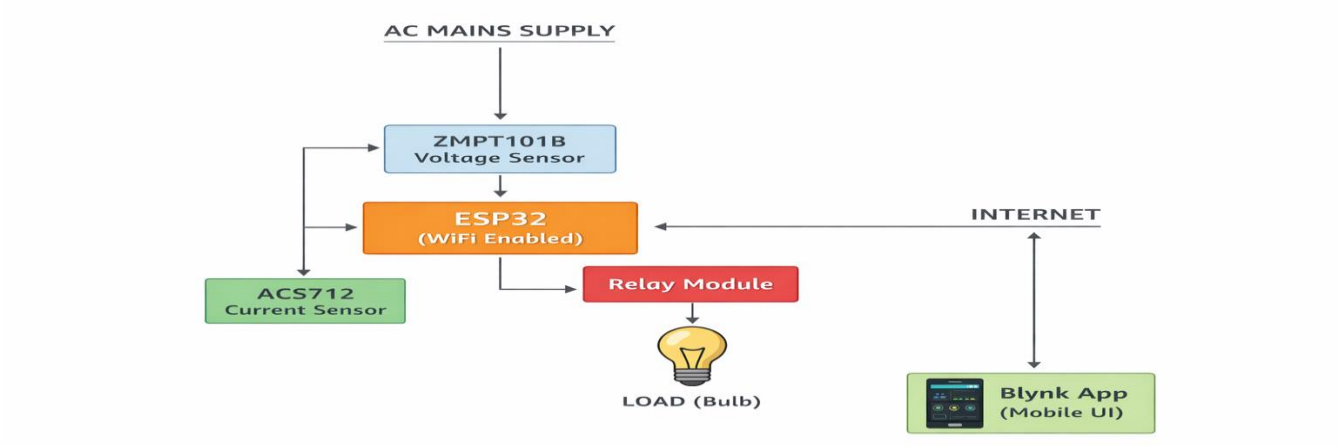


Figure 1 : Block Diagram of Model

Working of the Design Model:

The smart energy meter system works by measuring electrical parameters and sending the data to a mobile application for monitoring and control.

1. Power Supply Input

The AC mains supply is given to the system. The load (bulb) is connected through the relay module.

2. Voltage Measurement

The ZMPT101B Voltage Sensor Module senses the AC voltage and provides a scaled analog signal to the ESP32.

3. Current Measurement

The ACS712 current sensor measures the current flowing through the load and sends an analog signal to the ESP32.

4. Data Processing (ESP32)

The ESP32 Development Board reads both sensor values using its ADC and calculates:

- Voltage (V)
- Current (I)
- Power using:

  P=V×I$P=V\times I$
- Energy consumption over time

5. IoT Communication

The ESP32 connects to Wi-Fi and sends real-time data to the Blynk app using virtual pins.

6. User Monitoring and Control

- The user can view voltage, current, power, and energy on the mobile app
- The user can turn the load ON/OFF using a button in the app

7. Relay Operation

Based on user command or system conditions (such as overload), the ESP32 controls the relay to switch the bulb ON or OFF.

Pictures of Prototype:

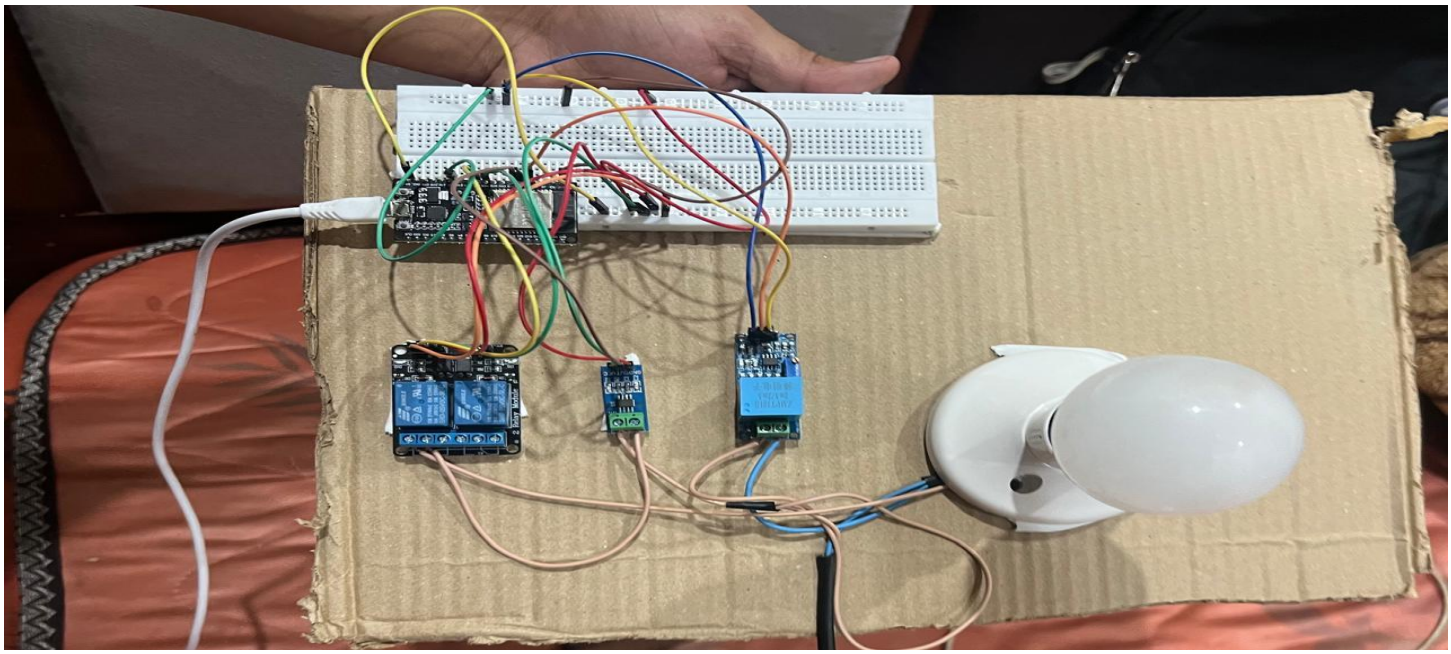
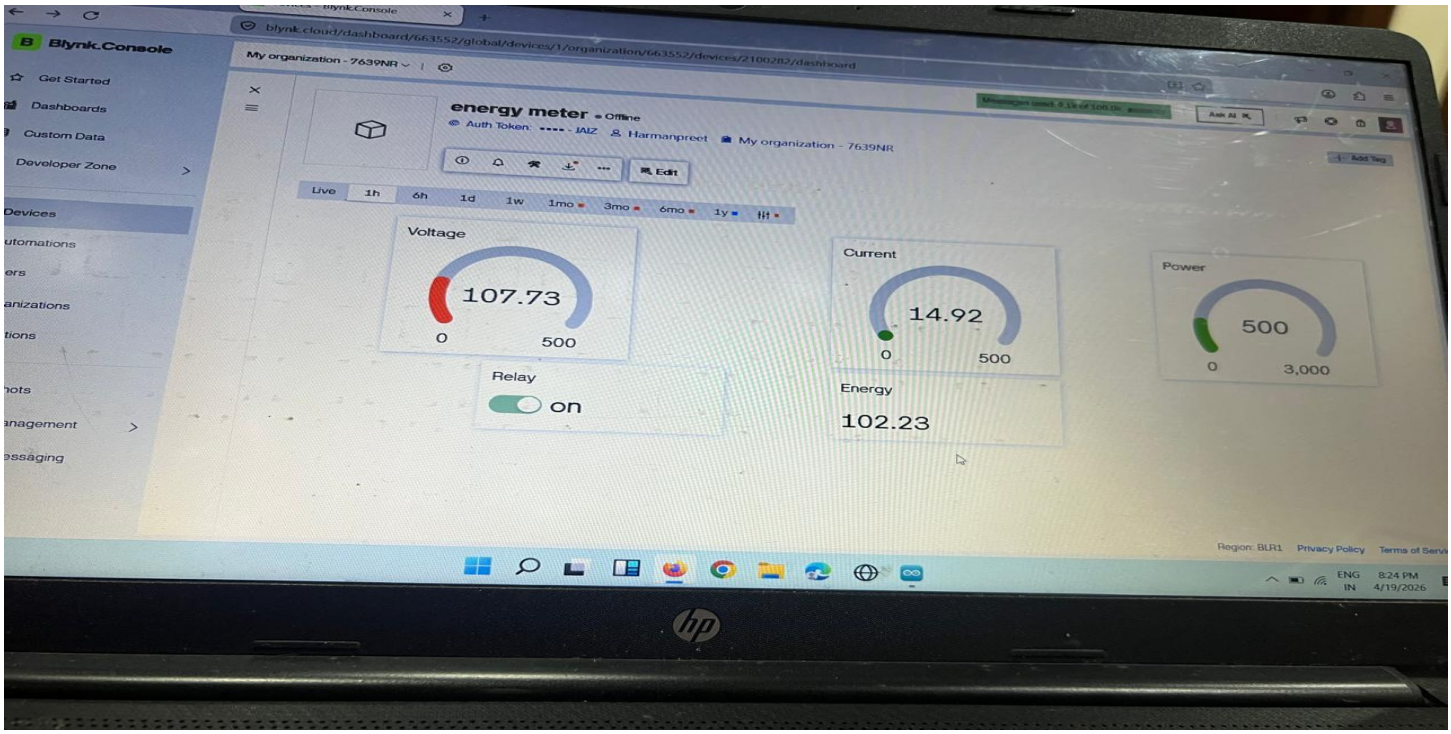


Figure 2: Designed Model

Output of Design Model/Prototype



Learning outcomes (What I have learnt):

1. **Understanding of Embedded Systems**
   Gained knowledge of working with the ESP32 Development Board for real-time data processing and control.
2. **Sensor Interfacing Skills**
   Learned how to interface and use sensors like the ZMPT101B Voltage Sensor Module and ACS712 for measuring electrical parameters.
3. **Knowledge of Energy Measurement**
   Understood how to calculate voltage, current, power, and energy consumption in practical applications.
4. **IoT Application Development**
   Developed skills in IoT by connecting the system to the Blynk for real-time monitoring and control.
5. **Practical Implementation of Smart Systems**
   Gained hands-on experience in designing and implementing a smart energy monitoring and load control system.